

10.5.6 Using a computer, compute and plot the Liapunov exponent as a function of r for the sine map $x_{n+1} = r \sin \pi x_n$, for $0 \leq x_n \leq 1$ and $0 \leq r \leq 1$.

10.5.7 The graph in Figure 10.5.2 suggests that $\lambda = 0$ at each period-doubling bifurcation value r_n . Show analytically that this is correct.

10.6 Universality and Experiments

The first two exercises deal with the sine map $x_{n+1} = r \sin \pi x_n$, where $0 < r \leq 1$ and $x \in [0,1]$. The goal is to learn about some of the practical problems that come up when one tries to estimate δ numerically.

10.6.1 (Naive approach)

- At each of 200 equally spaced r values, plot x_{700} through x_{1000} vertically above r , starting from some random initial condition x_0 . Check your orbit diagram against Figure 10.6.2 to be sure your program is working.
- Now go to finer resolution near the period-doubling bifurcations, and estimate r_n , for $n = 1, 2, \dots, 6$. Try to achieve five significant figures of accuracy.
- Use the numbers from (b) to estimate the Feigenbaum ratio $\frac{r_n - r_{n-1}}{r_{n+1} - r_n}$.

(Note: To get accurate estimates in part (b), you need to be clever, or careful, or both. As you probably found, a straightforward approach is hampered by “critical slowing down”—the convergence to a cycle becomes unbearably slow when that cycle is on the verge of period-doubling. This makes it hard to decide precisely where the bifurcation occurs. To achieve the desired accuracy, you may have to use double precision arithmetic, and about 10^4 iterates. But maybe you can find a shortcut by reformulating the problem.)

10.6.2 (Superstable cycles to the rescue) The “critical slowing down” encountered in the previous problem is avoided if we compute R_n instead of r_n . Here R_n denotes the value of r at which the sine map has a superstable cycle of period 2^n .

- Explain why it should be possible to compute R_n more easily and accurately than r_n .
- Compute the first six R_n ’s and use them to estimate δ .

If you’re interested in knowing the *best* way to compute δ , see Briggs (1991) for the state of the art.

10.6.3 (Qualitative universality of patterns) The U-sequence dictates the ordering of the windows, but it actually says more: it dictates the *iteration pattern* within each window. (See Exercise 10.4.7 for the definition of iteration patterns.) For instance, consider the large period-6 window for the logistic and sine maps, visible in Figure 10.6.2.

- For both maps, plot the cobweb for the corresponding superstable 6-cycle, given that it occurs at $r = 3.6275575$ for the logistic map and $r = 0.8811406$ for the sine map. (This cycle acts as a representative for the whole window.)
- Find the iteration pattern for both cycles, and confirm that they match.

10.6.4 (Period 4) Consider the iteration patterns of all possible period-4 orbits for the logistic map, or any other unimodal map governed by the U-sequence.

- Show that only two patterns are possible for period-4 orbits: RLL and RLR .
- Show that the period-4 orbit with pattern RLL always occurs after RLR , i.e., at a larger value of r .

10.6.5 (Unfamiliar later cycles) The final superstable cycles of periods 5, 6, 4, 6, 5, 6 in the logistic map occur at approximately the following values of r : 3.9057065, 3.9375364, 3.9602701, 3.9777664, 3.9902670, 3.9975831 (Metropolis et al. 1973). Notice that they're all near the end of the orbit diagram. They have tiny windows around them and tend to be overlooked.

- Plot the cobwebs for these cycles.
- Did you find it hard to obtain the cycles of periods 5 and 6? If so, can you explain why this trouble occurred?

10.6.6 (A trick for locating superstable cycles) Hao and Zheng (1989) give an amusing algorithm for finding a superstable cycle with a specified iteration pattern. The idea works for any unimodal map, but for convenience, consider the map $x_{n+1} = r - x_n^2$, for $0 \leq r \leq 2$. Define two functions $R(y) = \sqrt{r - y}$, $L(y) = -\sqrt{r - y}$. These are the right and left branches of the inverse map.

- For instance, suppose we want to find the r corresponding to the superstable 5-cycle with pattern $RLLR$. Then Hao and Zheng show that this amounts to solving the equation $r = RLLR(0)$. Show that when this equation is written out explicitly, it becomes

$$r = \sqrt{r + \sqrt{r + \sqrt{r - \sqrt{r}}}}.$$

- Solve this equation numerically by iterating the map

$$r_{n+1} = \sqrt{r_n + \sqrt{r_n + \sqrt{r_n - \sqrt{r_n}}}},$$

starting from any reasonable guess, e.g., $r_0 = 2$. Show numerically that r_n converges rapidly to 1.860782522. . . .

- Verify that the answer to (b) yields a cycle with the desired pattern.

10.7 Renormalization

10.7.1 (Hands on the functional equation) The functional equation $g(x) = \alpha g^2(x/\alpha)$ arose in our renormalization analysis of period-doubling. Let's

approximate its solution by brute force, assuming that $g(x)$ is even and has a quadratic maximum at $x = 0$.

- Suppose $g(x) \approx 1 + c_2 x^2$ for small x . Solve for c_2 and α . (Neglect $O(x^4)$ terms.)
- Now assume $g(x) \approx 1 + c_2 x^2 + c_4 x^4$, and use Mathematica, Maple, Macsyma (or hand calculation) to solve for α, c_2, c_4 . Compare your approximate results to the “exact” values $\alpha \approx -2.5029 \dots, c_2 \approx -1.527 \dots, c_4 \approx 0.1048 \dots$.

10.7.2 Given a map $y_{n+1} = f(y_n)$, rewrite the map in terms of a rescaled variable $x_n = \alpha y_n$. Use this to show that rescaling and inversion converts $f^2(x, R_1)$ into $\alpha f^2(x/\alpha, R_1)$, as claimed in the text.

10.7.3 Show that if g is a solution of the functional equation, so is $\mu g(x/\mu)$, with the same α .

10.7.4 (Wildness of the universal function $g(x)$) Near the origin $g(x)$ is roughly parabolic, but elsewhere it must be rather wild. In fact, the function $g(x)$ has infinitely many wiggles as x ranges over the real line. Verify these statements by demonstrating that $g(x)$ crosses the lines $y = \pm x$ infinitely many times. (Hint: Show that if x^* is a fixed point of $g(x)$, then so is αx^* .)

10.7.5 (Crudest possible estimate of α) Let $f(x, r) = r - x^2$.

- Write down explicit expressions for $f(x, R_0)$ and $\alpha f^2(x/\alpha, R_1)$.
- The two functions in (a) are supposed to resemble each other near the origin, if α is chosen correctly. (That’s the idea behind Figure 10.7.3.) Show the $O(x^2)$ coefficients of the two functions agree if $\alpha = -2$.

10.7.6 (Improved estimate of α) Redo Exercise 10.7.5 to one higher order: Let $f(x, r) = r - x^2$ again, but now compare $\alpha f^2(x/\alpha, R_1)$ to $\alpha^2 f^4(x/\alpha^2, R_2)$ and match the coefficients of the lowest powers of x . What value of α is obtained in this way?

10.7.7 (Quartic maxima) Develop the renormalization theory for functions with a *fourth-degree* maximum, e.g., $f(x, r) = r - x^4$. What approximate value of α is predicted by the methods of Exercises 10.7.1 and 10.7.5? Estimate the first few terms in the power series for the universal function $g(x)$. By numerical experimentation, estimate the new value of δ for the quartic case.

See Briggs (1991) for precise values of α and δ for this fourth-degree case, as well as for all other integer degrees between 2 and 12.

10.7.8 (Renormalization approach to intermittency: algebraic version) Consider the map $x_{n+1} = f(x_n, r)$, where $f(x_n, r) = -r + x - x^2$. This is the normal form for any map close to a tangent bifurcation.

- Show that the map undergoes a tangent bifurcation at the origin when $r = 0$.
- Suppose r is small and positive. By drawing a cobweb, show that a typical orbit takes many iterations to pass through the bottleneck at the origin.

- c) Let $N(r)$ denote the typical number of iterations of f required for an orbit to get through the bottleneck. Our goal is to see how $N(r)$ scales with r as $r \rightarrow 0$. We use a renormalization idea: Near the origin, f^2 looks like a rescaled version of f , and hence it too has a bottleneck there. Show that it takes approximately $\frac{1}{2}N(r)$ iterations for orbits of f^2 to pass through the bottleneck.
- d) Expand $f^2(x, r)$ and keep only the terms through $O(x^2)$. Rescale x and r to put this new map into the desired normal form $F(X, R) \approx -R + X - X^2$. Show that this renormalization implies the recursive relation

$$\frac{1}{2}N(r) \approx N(4r).$$

- e) Show that the equation in (d) has solutions $N(r) = ar^b$ and solve for b .

10.7.9 (Renormalization approach to intermittency: functional version) Show that if the renormalization procedure in Exercise 10.7.8 is done exactly, we are led to the functional equation

$$g(x) = \alpha g^2(x/\alpha)$$

(just as in the case of period-doubling!) but with new boundary conditions appropriate to the tangent bifurcation:

$$g(0) = 0, \quad g'(0) = 1.$$

Unlike the period-doubling case, this functional equation can be solved *explicitly* (Hirsch et al. 1982).

- a) Verify that a solution is $\alpha = 2$, $g(x) = x/(1 + ax)$, with a arbitrary.
- b) Explain why $\alpha = 2$ is almost obvious, in retrospect. (Hint: Draw cobwebs for both g and g^2 for an orbit passing through the bottleneck. Both cobwebs look like staircases; compare the lengths of their steps.)

10.7.10 Fill in the missing algebraic steps in the concrete renormalization calculation for period-doubling. Let $f(x) = -(1 + \mu)x + x^2$. Expand $p + \eta_{n+1} = f^2(p + \eta_n)$ in powers of the small deviation η_n , using the fact that p is a fixed point of f^2 . Thereby confirm that (10.7.4) and (10.7.5) are correct.

10.7.11 Give a cobweb analysis of (10.7.10), starting from the initial condition $\mu_1 = 0$. Show that $\mu_k \rightarrow \mu^*$, where $\mu^* > 0$ is a stable fixed point corresponding to the onset of chaos.

FRACTALS

11.0 Introduction

Back in Chapter 9, we found that the solutions of the Lorenz equations settle down to a complicated set in phase space. This set is the strange attractor. As Lorenz (1963) realized, the geometry of this set must be very peculiar, something like an “infinite complex of surfaces.” In this chapter we develop the ideas needed to describe such strange sets more precisely. The tools come from fractal geometry.

Roughly speaking, *fractals* are complex geometric shapes with fine structure at arbitrarily small scales. Usually they have some degree of self-similarity. In other words, if we magnify a tiny part of a fractal, we will see features reminiscent of the whole. Sometimes the similarity is exact; more often it is only approximate or statistical.

Fractals are of great interest because of their exquisite combination of beauty, complexity, and endless structure. They are reminiscent of natural objects like mountains, clouds, coastlines, blood vessel networks, and even broccoli, in a way that classical shapes like cones and squares can’t match. They have also turned out to be useful in scientific applications ranging from computer graphics and image compression to the structural mechanics of cracks and the fluid mechanics of viscous fingering.

Our goals in this chapter are modest. We want to become familiar with the simplest fractals and to understand the various notions of fractal dimension. These ideas will be used in Chapter 12 to clarify the geometric structure of strange attractors.

Unfortunately, we will not be able to delve into the scientific applications of fractals, nor the lovely mathematical theory behind them. For the clearest introduction to the theory and applications of fractals, see Falconer (1990). The books of Mandelbrot (1982), Peitgen and Richter (1986), Barnsley (1988), Feder (1988), and Schroeder (1991) are also recommended for their many fascinating pictures and examples.

11.1 Countable and Uncountable Sets

This section reviews the parts of set theory that we'll need in later discussions of fractals. You may be familiar with this material already; if not, read on.

Are some infinities larger than others? Surprisingly, the answer is yes. In the late 1800s, Georg Cantor invented a clever way to compare different infinite sets. Two sets X and Y are said to have the same *cardinality* (or number of elements) if there is an invertible mapping that pairs each element $x \in X$ with precisely one $y \in Y$. Such a mapping is called a **one-to-one correspondence**; it's like a buddy system, where every x has a buddy y , and no one in either set is left out or counted twice.

A familiar infinite set is the set of natural numbers $\mathbf{N} = \{1, 2, 3, 4, \dots\}$. This set provides a basis for comparison—if another set X can be put into one-to-one correspondence with the natural numbers, then X is said to be **countable**. Otherwise X is **uncountable**.

These definitions lead to some surprising conclusions, as the following examples show.

EXAMPLE 11.1.1:

Show that the set of even natural numbers $E = \{2, 4, 6, \dots\}$ is countable.

Solution: We need to find a one-to-one correspondence between E and $\mathbf{N}$. Such a correspondence is given by the invertible mapping that pairs each natural number n with the even number $2n$; thus $1 \leftrightarrow 2$, $2 \leftrightarrow 4$, $3 \leftrightarrow 6$, and so on.

Hence there are exactly as many even numbers as natural numbers. You might have thought that there would be only *half as* many, since all the odd numbers are missing! ■

There is an equivalent characterization of countable sets which is frequently useful. A set X is countable if it can be written as a list $\{x_1, x_2, x_3, \dots\}$, with every $x \in X$ appearing somewhere in the list. In other words, given any x , there is some finite n such that $x_n = x$.

A convenient way to exhibit such a list is to give an algorithm that systematically counts the elements of X . This strategy is used in the next two examples.

EXAMPLE 11.1.2:

Show that the integers are countable.

Solution: Here's an algorithm for listing all the integers: We start with 0 and then work in order of increasing absolute value. Thus the list is $\{0, 1, -1, 2, -2, 3, -3, \dots\}$. Any particular integer appears eventually, so the integers are countable. ■

EXAMPLE 11.1.3:

Show that the positive rational numbers are countable.

Solution: Here's a *wrong* way: we start listing the numbers $\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \dots$ in order. Unfortunately we never finish the $\frac{1}{n}$'s and so numbers like $\frac{2}{3}$ are never counted!

The right way is to make a table where the pq -th entry is p/q . Then the rationals can be counted by the weaving procedure shown in Figure 11.1.1. Any given p/q is reached after a finite number of steps, so the rationals are countable. ■

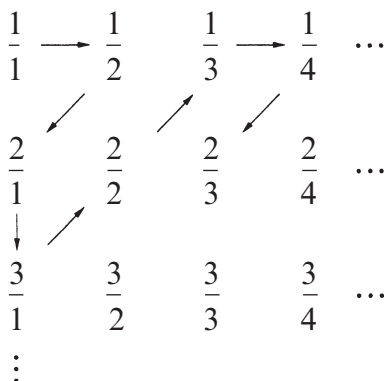


Figure 11.1.1

Now we consider our first example of an uncountable set.

EXAMPLE 11.1.4:

Let X denote the set of all real numbers between 0 and 1. Show that X is uncountable.

Solution: The proof is by contradiction. If X were countable, we could list all the real numbers between 0 and 1 as a set $\{x_1, x_2, x_3, \dots\}$. Rewrite these numbers in decimal form:

$$\begin{aligned}x_1 &= 0.x_{11}x_{12}x_{13}x_{14}\dots \\x_2 &= 0.x_{21}x_{22}x_{23}x_{24}\dots \\x_3 &= 0.x_{31}x_{32}x_{33}x_{34}\dots \\&\vdots\end{aligned}$$

where x_{ij} denotes the j th digit of the real number x_i .

To obtain a contradiction, we'll show that there's a number r between 0 and 1 that is *not* on the list. Hence any list is necessarily incomplete, and so the reals are uncountable.

We construct r as follows: its first digit is *anything other than* x_{11} , the first digit of x_1 . Similarly, its second digit is anything other than the second digit of x_2 . In general, the n th digit of r is $\bar{x}_{nm}$, defined as any digit other than x_{nm} . Then we claim that the number $r = \bar{x}_{11}\bar{x}_{22}\bar{x}_{33}\dots$ is not on the list. Why not? It can't be equal to x_1 , because it differs from x_1 in the first decimal place. Similarly, r differs from x_2 in the second decimal place, from x_3 in the third decimal place, and so on. Hence r is not on the list, and thus X is uncountable. ■

This argument (devised by Cantor) is called the **diagonal argument**, because r is constructed by changing the diagonal entries x_{nn} in the matrix of digits $[x_{ij}]$.

11.2 Cantor Set

Now we turn to another of Cantor's creations, a fractal known as the Cantor set. It is simple and therefore pedagogically useful, but it is also much more than that—as we'll see in Chapter 12, the Cantor set is intimately related to the geometry of strange attractors.

Figure 11.2.1 shows how to construct the Cantor set.

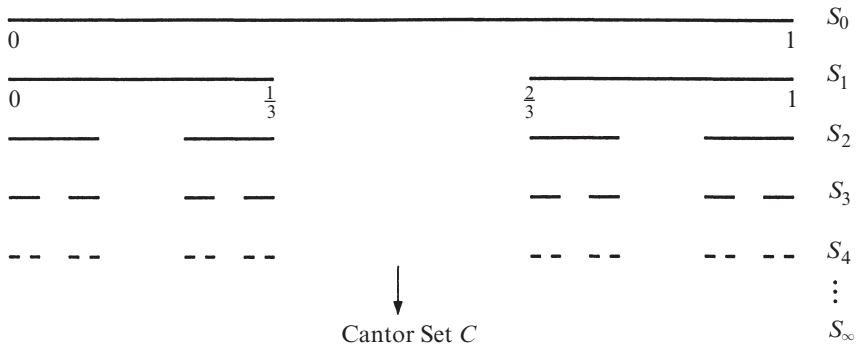


Figure 11.2.1

We start with the closed interval $S_0 = [0, 1]$ and remove its open middle third, i.e., we delete the interval $(\frac{1}{3}, \frac{2}{3})$ and leave the endpoints behind. This produces the pair of closed intervals shown as S_1 . Then we remove the open middle thirds of *those* two intervals to produce S_2 , and so on. The limiting set $C = S_\infty$ is the **Cantor set**. It is difficult to visualize, but Figure 11.2.1 suggests that it consists of an infinite number of infinitesimal pieces, separated by gaps of various sizes.

Fractal Properties of the Cantor Set

The Cantor set C has several properties that are typical of fractals more generally:

1. *C has structure at arbitrarily small scales.* If we enlarge part of C repeatedly, we continue to see a complex pattern of points separated by gaps of various sizes. This structure is neverending, like worlds within worlds. In contrast, when we look at a smooth curve or surface under repeated magnification, the picture becomes more and more featureless.
2. *C is self-similar.* It contains smaller copies of itself at all scales. For instance, if we take the left part of C (the part contained in the interval $[0, \frac{1}{3}]$) and enlarge it by a factor of three, we get C back again. Similarly, the parts of C in each of the four intervals of S_2 are geometrically similar to C , except scaled down by a factor of nine.

If you're having trouble seeing the self-similarity, it may help to think about the sets S_n rather than the mind-boggling set S_∞ . Focus on the left half of S_2 —it looks just like S_1 , except three times smaller. Similarly, the left half of S_3 is S_2 , reduced by a factor of three. In general, the left half of S_{n+1} looks like *all* of S_n , scaled down by three. Now set $n = \infty$. The conclusion is that the left half of S_∞ looks like S_∞ , scaled down by three, just as we claimed earlier.

Warning: The strict self-similarity of the Cantor set is found only in the simplest fractals. More general fractals are only approximately self-similar.

3. *The dimension of C is not an integer.* As we'll show in Section 11.3, its dimension is actually $\ln 2 / \ln 3 \approx 0.63$! The idea of a noninteger dimension is bewildering at first, but it turns out to be a natural generalization of our intuitive ideas about dimension, and provides a very useful tool for quantifying the structure of fractals.

Two other properties of the Cantor set are worth noting, although they are not fractal properties as such: *C has measure zero* and *it consists of uncountably many points*. These properties are clarified in the examples below.

EXAMPLE 11.2.1:

Show that the *measure* of the Cantor set is zero, in the sense that it can be covered by intervals whose total length is arbitrarily small.

Solution: Figure 11.2.1 shows that each set S_n completely covers all the sets that come after it in the construction. Hence the Cantor set $C = S_\infty$ is covered by *each* of the sets S_n . So the total length of the Cantor set must be less than the total length of S_n , for any n . Let L_n denote the length of S_n . Then from Figure 11.2.1 we see that $L_0 = 1$, $L_1 = \frac{2}{3}$, $L_2 = (\frac{2}{3})(\frac{2}{3}) = (\frac{2}{3})^2$, and in general, $L_n = (\frac{2}{3})^n$. Since $L_n \rightarrow 0$ as $n \rightarrow \infty$, the Cantor set has a total length of zero. ■

Example 11.2.1 suggests that the Cantor set is “small” in some sense. On the other hand, it contains tremendously many points—uncountably many, in fact. To see this, we first develop an elegant characterization of the Cantor set.

EXAMPLE 11.2.2:

Show that the Cantor set C consists of all points $c \in [0,1]$ that have no 1's in their base-3 expansion.

Solution: The idea of expanding numbers in different bases may be unfamiliar, unless you were one of those children who was taught “New Math” in elementary school. Now you finally get to see why base-3 is useful!

First let's remember how to write an arbitrary number $x \in [0,1]$ in base-3. We expand in powers of $1/3$: thus if $x = \frac{a_1}{3} + \frac{a_2}{3^2} + \frac{a_3}{3^3} + \dots$, then $x = .a_1a_2a_3\dots$ in base-3, where the digits a_n are 0, 1, or 2. This expansion has a nice geometric interpretation (Figure 11.2.2).

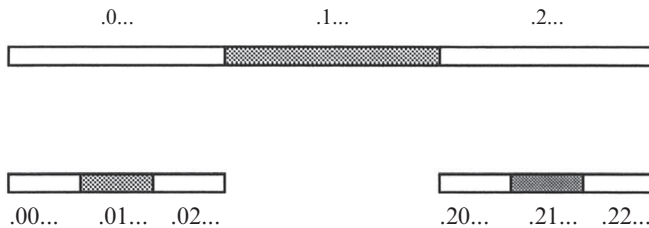


Figure 11.2.2

If we imagine that $[0,1]$ is divided into three equal pieces, then the first digit a_1 tells us whether x is in the left, middle, or right piece. For instance, all numbers with $a_1 = 0$ are in the left piece. (Ordinary base-10 works the same way, except that we divide $[0,1]$ into ten pieces instead of three.) The second digit a_2 provides more refined information: it tells us whether x is in the left, middle, or right third of a given piece. For instance, points of the form $x = .01\dots$ are in the middle part of the left third of $[0,1]$, as shown in Figure 11.2.2.

Now think about the base-3 expansion of points in the Cantor set C . We deleted the middle third of $[0,1]$ at the first stage of constructing C ; this removed all points whose first digit is 1. So those points can't be in C . The points left over (the only ones with a chance of ultimately being in C) must have 0 or 2 as their first digit. Similarly, points whose *second* digit is 1 were deleted at the next stage in the construction. By repeating this argument, we see that C consists of all points whose base-3 expansion contains no 1's, as claimed. ■

There's still a fussy point to be addressed. What about endpoints like $\frac{1}{3} = .1000\dots$? It's in the Cantor set, yet it has a 1 in its base-3 expansion. Does this contradict what we said above? No, because this point can also be written solely in terms of 0's and 2's, as follows: $\frac{1}{3} = .1000\dots = .02222\dots$. By this trick, each point in the Cantor set can be written such that no 1's appear in its base-3 expansion, as claimed.

Now for the payoff.

EXAMPLE 11.2.3:

Show that the Cantor set is uncountable.

Solution: This is just a rewrite of the Cantor diagonal argument of Example 11.1.4, so we'll be brief. Suppose there were a list $\{c_1, c_2, c_3, \dots\}$ of all points in C . To show that C is uncountable, we produce a point $\bar{c}$ that is in C but not on the list. Let c_{ij} denote the j th digit in the base-3 expansion of c_i . Define $\bar{c} = \bar{c}_{11}\bar{c}_{22}\dots$, where the overbar means we switch 0's and 2's: thus $\bar{c}_{nn} = 0$ if $c_{nn} = 2$ and $\bar{c}_{nn} = 2$ if $c_{nn} = 0$. Then $\bar{c}$ is in C , since it's written solely with 0's and 2's, but $\bar{c}$ is not on the list, since it differs from c_n in the n th digit. This contradicts the original assumption that the list is complete. Hence C is uncountable. ■

11.3 Dimension of Self-Similar Fractals

What is the “dimension” of a set of points? For familiar geometric objects, the answer is clear—lines and smooth curves are one-dimensional, planes and smooth surfaces are two-dimensional, solids are three-dimensional, and so on. If forced to give a definition, we could say that *the dimension is the minimum number of coordinates needed to describe every point in the set*. For instance, a smooth curve is one-dimensional because every point on it is determined by one number, the arc length from some fixed reference point on the curve.

But when we try to apply this definition to fractals, we quickly run into paradoxes. Consider the **von Koch curve**, defined recursively in Figure 11.3.1.

We start with a line segment S_0 . To generate S_1 , we delete the middle third of S_0 and replace it with the other two sides of an equilateral triangle. Subsequent stages are generated recursively by the same rule: S_n is obtained by replacing the middle third of each line segment in S_{n-1} by the other two sides of an equilateral triangle. The limiting set $K = S_\infty$ is the von Koch curve.

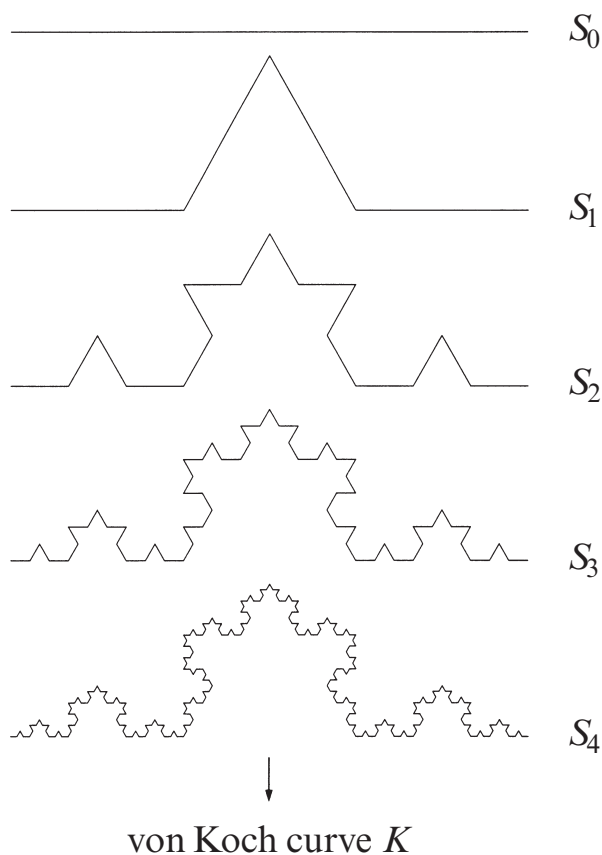


Figure 11.3.1

A Paradox

What is the dimension of the von Koch curve? Since it's a curve, you might be tempted to say it's one-dimensional. But the trouble is that K has *infinite arc length*! To see this, observe that if the length of S_0 is L_0 , then the length of S_1 is $L_1 = \frac{4}{3}L_0$, because S_1 contains four segments, each of length $\frac{1}{3}L_0$. The length increases by a factor of $\frac{4}{3}$ at each stage of the construction, so $L_n = (\frac{4}{3})^n L_0 \rightarrow \infty$ as $n \rightarrow \infty$.

Moreover, the arc length between *any* two points on K is infinite, by similar reasoning. Hence points on K aren't determined by their arc length from a particular point, because every point is infinitely far from every other!

This suggests that K is more than one-dimensional. But would we really want to say that K is two-dimensional? It certainly doesn't seem to have any "area." So the dimension should be *between* 1 and 2, whatever that means.

With this paradox as motivation, we now consider some improved notions of dimension that can cope with fractals.